# Multiply and divide by powers of ten

I can... explain place-value patterns when scaling by ten or one hundred.

## Each digit takes a new value

$3.45 \times 10 = 34.5$

$3.45 \times 100 = 345$

## Make the value smaller

$3.45 \div 10 = 0.345$

$3.45 \times 0.1 = 0.345$

## Explain the digits

In 3.45, the 3 means 3 ones. After multiplying by 10, it means 3 tens. The value changed; we did not simply attach a zero.